

ECON108 Winter 2025

Milestone 3

Seeun Park

Research Question: I examine how racial stereotypes and political views influence individuals' perceptions of welfare spending.

Question 1. Model building

Model 1: $\Pr(Welfare_i = 1) = \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + u_i$ (1)

$Welfare_i$ is a binary variable that takes the value of 1 if individual i thinks the government's welfare spending is too much, and 0 otherwise.

Model 2: $\Pr(Welfare_i = 1) = \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i + \beta_4 \text{Conservative}_i + u_i$ (2)

In Model 2, I add political view dummies because political views may influence individuals' perceptions of specific races' work ethic as well as opinions on welfare spending. Omitting these variables could bias my estimates. Column (2) shows that controlling for political views slightly reduces the magnitude of the coefficients on the variables of interest, **White lazy** and **Black lazy**, indicating that the previous estimates in Model 1 capture the relationship between political views and the outcome variable. Additionally, because the political view dummies are statistically significant at the 5% level, I prefer Model 2 to Model 1.

Model 3: $\Pr(Welfare_i = 1) = \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i + \beta_4 \text{Conservative}_i + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i + \beta_9 \text{EduYears}_i + \beta_{10} \text{Year}_t + u_i$ (3)

In the second adjustment, I control for demographic characteristics that may influence both the variables of interest and the outcome variable. Adding these variables makes the coefficient on **White Lazy** statistically insignificant, suggesting that the previous estimate was driven by omitted demographic characteristics. Therefore, I prefer Model 3 over Model 2.

Model 4: $\Pr(Welfare_i = 1) = \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i + \beta_4 \text{Conservative}_i + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i + \beta_9 \text{EduYears}_i + \sum_t \gamma_t \text{Year}_t + u_i$ (4)

,where $t \in \{1994, 1996, \dots, 2016, 2018, 2021, 2022\}$ and 1990 is omitted

In Model 4, I control for year fixed effects instead of a linear time trend. A linear time trend assumes that the effect of time on the outcome variable is constant and changes at a uniform rate across all years. However, this may not be the case, especially during events like the Great Recession or the COVID-19 pandemic, when people's perceptions of welfare spending may have differed from other years. To capture unobserved factors that may differ across years but remain constant within individuals, I include year fixed effects, with 1990 as the baseline year. Changing from a linear time trend to year fixed effects does not change the results significantly, but it still means that year fixed effects capture some unobserved factors that may cause omitted variable bias. Therefore, I prefer Model 4 to Model 3.

$$\begin{aligned}
 \textbf{Model 5: } \Pr(Welfare_i = 1) = & \Phi\left(\beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i \right. \\
 & + \beta_4 \text{Conservative}_i + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i \\
 & \left. + \beta_9 \text{EduYears}_i + \sum_{t=1}^T \gamma_t \text{Year}_t + u_i\right) \quad (5) \\
 & , \text{where } t \in \{1990, 1994, 1996, \dots, 2016, 2018, 2021, 2022\}
 \end{aligned}$$

When I check the predicted outcomes from Model 4, I find that a few observations fall outside the [0,1] range. Therefore, in Model 5, I estimate the same model using a probit specification. I prefer Model 4 for three reasons. First, while the size of the estimates changes slightly, the overall pattern remains consistent with Model 4. Second, the count R-squared values for the two models are almost identical, at 0.798. Third, less than 10% of the predicted values fall below zero, with the minimum value being -0.002, which is very close to zero. Considering the ease of interpretation and the limited gain from using the probit model, I prefer Model 4.

The Final Preferred Model (Model 4)

$$\begin{aligned}
 \Pr(Welfare_i = 1) = & \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i + \beta_4 \text{Conservative}_i \\
 & + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i + \beta_9 \text{EduYears}_i \\
 & + \sum_t \gamma_t \text{Year}_t + u_i \quad , \text{where } t \in \{1994, 1996, \dots, 2016, 2018, 2021, 2022\}
 \end{aligned}$$

Question 2. Formatted table of regression output**Outcome:** Welfare Spending is Too Much (=1)

	(1)	(2)	(3)	(4)	(5)
	LPM	LPM	LPM	LPM	Probit
White lazy	-0.0534*** (0.0137)	-0.0407*** (0.0137)	-0.0134 (0.0139)	-0.0112 (0.0138)	-0.0109 (0.0138)
Black lazy	0.143*** (0.00973)	0.119*** (0.00970)	0.107*** (0.0100)	0.105*** (0.00994)	0.102*** (0.00967)
Liberal		-0.140*** (0.0109)	-0.139*** (0.0110)	-0.135*** (0.0110)	-0.139*** (0.0110)
Conservative		0.135*** (0.0104)	0.122*** (0.0106)	0.124*** (0.0105)	0.116*** (0.0101)
Female			-0.000760 (0.00891)	-0.00154 (0.00886)	-0.00116 (0.00885)
Black			-0.140*** (0.0131)	-0.143*** (0.0130)	-0.145*** (0.0133)
Other Race			-0.117*** (0.0162)	-0.113*** (0.0161)	-0.112*** (0.0163)
Age			-0.000637** (0.000263)	-0.000545** (0.000262)	-0.000542** (0.000261)
Years of Education			0.00214 (0.00155)	0.00224 (0.00155)	0.00211 (0.00155)
Year			-0.00325*** (0.000465)		
Constant	0.380*** (0.00547)	0.382*** (0.00790)	6.949*** (0.931)	0.378*** (0.0323)	
Observations	12,134	11,776	11,522	11,522	11,522
R-squared	0.018	0.066	0.082	0.097	
Year FE				Y	Y

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Question 3. Discussion of final model*Qualitative variables*

- **White lazy:** Individuals who perceive White people as lazy are 0.0112 (or 1.12 percentage points) less likely to think the government is spending too much on welfare, compared to those who do not perceive them as lazy. However, this estimate is not statistically significant at the 5% level.
- **Black lazy:** The probability that individuals think the government is spending too much on welfare is 10.5 percentage points higher for those who perceive Black people as lazy, compared to those who do not. This estimate is statistically significant at the 5% level.
- **Liberal:** Compared to individuals with moderate political views, liberals are 13.5 percentage points less likely to think that welfare spending is excessive, holding all else constant. This estimate is statistically significant at the 5% level.
- **Conservative:** Conservatives are 12.4 percentage points more likely to think that welfare spending is excessive, compared to individuals with moderate political views, holding all else constant. This estimate is statistically significant at the 5% level.
- **Female:** Females are 0.154 percentage points more likely to think that welfare spending is excessive, compared to males, holding all else constant. However, this estimate is not statistically significant at the 5% level.
- **Black:** Blacks are 14.3 percentage points less likely to think that the government spends too much on welfare, compared to Whites, holding all else constant. This estimate is statistically significant at the 5% level.
- **Other Race:** Individuals of other races are 11.3 percentage points less likely to think that the government spends too much on welfare, compared to Whites, holding all else constant. This estimate is statistically significant at the 5% level.

To summarize, the results indicate that racial stereotypes, an individual's race, and political views are closely related to perceptions of government spending on welfare. This suggests that politicians who hold biased views about Black people's work ethics, are more conservative, and are White, are more likely to believe that government spending on welfare is excessive and seek to reduce the welfare budget, which could widen economic inequality. This finding is especially pertinent in the current political climate, where a conservative White president with racially biased views has taken office in the United States.

Quantitative variables

- **Age:** For each additional year of age, the probability that an individual believes welfare spending is too much decreases by 0.05 percentage points, holding all other factors constant. Although statistically significant, this estimate is very close to zero.
- **Years of Education:** For each additional year of education, the probability that an individual believes welfare spending is too much increases by 0.2 percentage points, holding all other factors constant. The estimate is statistically significant at the 5% level.

Individuals with a higher level of education are more likely to think that the government's spending on welfare is excessive, while age is not related to this perception. Given that years of education are generally positively correlated with income, this finding is somewhat predictable.

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1  *=====*
2  * Econ 108 Winter 2025
3  * Milestone 3 (Seeun Park)
4  * 2/23/2025
5  *=====*
6
7  clear
8  set more off
9
10 cd "/Users/seeunpark/Dropbox/UCR/2024-2025/Winter2025/Econ108_W25/2_Lab/Econ108_W25_wd/Lab_GSSProject"
11
12 cap log close
13 log using Log_Econ108_W25_MS3, text replace
14
15 use gss_data_spark_clean, clear
16
17 *-----*
18 ** Model 1: initial model
19 reg welfare_much white_lazy black_lazy
20     outreg2 using "Econ108_W25_MS3_RegTable", excel replace
21
22 *-----*
23 ** Model 2: first adjustment (add more controls)
24 reg welfare_much white_lazy black_lazy pol3_lib pol3_cons
25     outreg2 using "Econ108_W25_MS3_RegTable", excel
26
27 *-----*
28 ** Model 3: second adjustment (add 5 common vars)
29 reg welfare_much white_lazy black_lazy pol3_lib pol3_cons female race_black race_other age educ year
30     outreg2 using "Econ108_W25_MS3_RegTable", excel
31
32
33 predict phat_lpm, xb
34     su phat_lpm
35 *Everything is between [0,1]
36
37 gen correct_lpm = 0
38     replace correct_lpm = 1 if phat_lpm > 0.5 & welfare_much == 1
39     replace correct_lpm = 1 if phat_lpm < 0.5 & welfare_much == 0
40     replace correct_lpm = . if phat_lpm == .
41     su correct_lpm
42
43 *-----*
44 ** Model 4: third adjustment (year FE)
45 reg welfare_much white_lazy black_lazy pol3_lib pol3_cons female race_black race_other age educ i.year
46     outreg2 using "Econ108_W25_MS3_RegTable", excel drop(i.year) addtext(Year FE, Y)
47
48
49 predict phat_lpm2, xb
50     su phat_lpm2
51 *Some are out of [0,1] range
52
53 gen correct_lpm2 = 0
54     replace correct_lpm2 = 1 if phat_lpm2 > 0.5 & welfare_much == 1
55     replace correct_lpm2 = 1 if phat_lpm2 < 0.5 & welfare_much == 0
56     replace correct_lpm2 = . if phat_lpm2 == .
57     su correct_lpm2
58
59 hist phat_lpm2, color(eltblue)
60 graph export fig_phat_lpm2.pdf, replace
61
62 *-----*
63 ** Model 5: fourth adjustment (probit)
64 *probit welfare_much white_lazy black_lazy pol3_lib pol3_cons female race_black race_other age educ i.year
65 * outreg2 using "Econ108_W25_MS3_RegTable", excel drop(i.year) addtext(Year FE, Y)
66
67 probit welfare_much white_lazy black_lazy pol3_lib pol3_cons female race_black race_other age educ i.year
68     margins, dydx(*) post
69     outreg2 using "Econ108_W25_MS3_RegTable", excel drop(i.year) addtext(Year FE, Y)
70
71
72 probit welfare_much white_lazy black_lazy pol3_lib pol3_cons female race_black race_other age educ i.year
73 predict phat_probit, xb
74
75 gen correct_probit = 0
76     replace correct_probit = 1 if phat_probit > 0.5 & welfare_much == 1
77     replace correct_probit = 1 if phat_probit < 0.5 & welfare_much == 0
78     replace correct_probit = . if phat_probit == .
79     su correct_probit
80
81
82 *considering that phat_lpm2's predicted values are mostly within the range and there is no gaining in terms of count R square,
83 I will choose LPM because of the easiness of interpretation (more intuitive and straightforward interpretation is possible)
84 */
85
86 log close

```